

CSE 4345: Big Data Analytics

Week 9, Part 2 — Research Article & Case Study

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Today's Agenda

- 1 Work 1 — Tufte (1983): The Visual Display of Quantitative Information
- 2 Work 2 — Shneiderman (1996): The Eyes Have It
- 3 Case Study — New York Times Graphics Desk (2010–2022)
- 4 Live Exercise

CLO Addressed

CLO3 — Explain data visualisation principles
PLO(b)

Seminal Works Covered

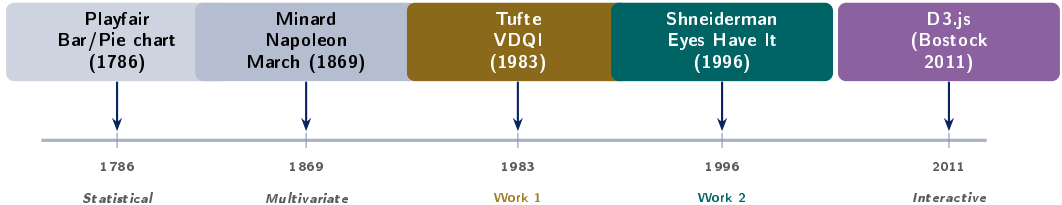
- 1 Tufte, E. R. (1983). *The Visual Display of Quantitative Information*. Graphics Press.
- 2 Shneiderman, B. (1996). *The Eyes Have It: A Task by Data Type Taxonomy for Information Visualizations*. IEEE Visual Languages.

Case Study

New York Times Graphics Desk (2010–2022): interactive data journalism using D3.js, reaching **6 million readers per day**; COVID-19 visualisations cited in policy briefs worldwide.

Work 1 — Tufte (1983): The Visual Display of Quantitative Information

Research Landscape: History of Data Visualisation



Why Tufte Matters for Big Data Visualisation

Big Data pipelines produce outputs humans must interpret. A chart with **chartjunk** on a 4K dashboard misleads as badly as a bad statistical test. Tufte's principles are the analytical foundation for every modern dashboard framework (Tableau, Grafana, D3). The week's D3 practicals are applications of Tufte's rules.

Tufte (1983): Publication Context

Full Citation

Tufte, E. R. (1983). **The Visual Display of Quantitative Information**. Graphics Press, Cheshire, CT. (2nd ed., 2001; 197 pp.)

Format: Book (self-published by Tufte; designed and typeset by Tufte himself)

Cited by: >20,000 (Google Scholar, 2024)

Author: Edward R. Tufte, Professor of Statistics, Graphic Design, and Political Economy (Yale University)

Note: the book was *rejected by every publisher* Tufte approached; he mortgaged his house to self-publish. It sold 40,000 copies in its first year.

Tufte's Central Claim

"Graphical excellence is that which gives to the viewer the greatest number of ideas in the shortest time with the least ink in the smallest space."

The Context: Why 1983?

- By 1983, computer-generated charts were proliferating in corporate reports and newspapers
- Most were produced by software that defaulted to heavy gridlines, 3-D effects, unnecessary colour, and excessive labelling
- Tufte analysed **4,000 charts** from science, business, and journalism publications to identify patterns of effective and deceptive design
- His contemporaries focused on *how to draw* charts; Tufte focused on *what to show* and *what to remove*

Technical Contribution 1 — Data-Ink Ratio & Chartjunk

The Data-Ink Ratio

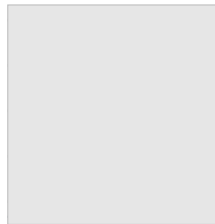
$$\text{Data-Ink Ratio} = \frac{\text{ink used to display data}}{\text{total ink used in the graphic}}$$

Goal: maximise data-ink ratio. Every non-data mark must justify its existence.

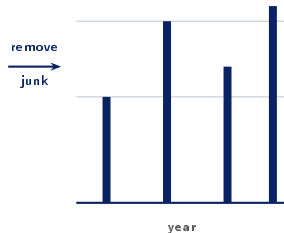
Chartjunk — Tufte's term for ink that does not convey data:

- **Vibrations:** optical illusions from cross-hatching and dense patterns
- **Grids:** heavy gridlines compete with data; use light or no gridlines
- **Ducks:** graphic elements that represent themselves rather than data (e.g., a chart drawn as a 3-D bar on a fake desktop)
- **3-D effects:** depth, shadows, and perspective distort quantitative comparisons — never use 3-D for 2-D data

LOW data-ink (chartjunk)



HIGH data-ink (Tufte style)



Technical Contribution 2 — Lie Factor, Small Multiples & Sparklines

The Lie Factor

$$\text{Lie Factor} = \frac{\text{size of effect shown in graphic}}{\text{size of effect in data}}$$

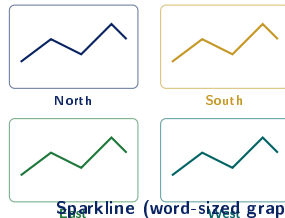
A lie factor of 1 is truthful. >1 : graphic *overstates* the effect. <1 : graphic *understates* it.

Classic example: a bar chart where the y-axis starts at 50 instead of 0. If the data effect is +10% but the bar appears to double in height, the lie factor is ≈ 10 .

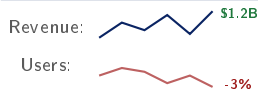
Small Multiples

The same chart structure repeated across many subsets of data.

Small Multiples (4 regions)



Sparkline (word-sized graphic)



Sparklines (Tufte 2006)

Sparklines are intense, simple, word-sized graphics embedded in text. Grafana and

Work 2 — Shneiderman (1996): The Eyes Have It

Shneiderman (1996): Publication Context & the Mantra

Full Citation

Shneiderman, B. (1996). **The Eyes Have It: A Task by Data Type Taxonomy for Information Visualizations.** In *Proceedings of the IEEE Symposium on Visual Languages*, pp. 336–343. Boulder, CO, USA.

Venue: IEEE Visual Languages — premier HCI/Viz venue

Cited by: >5,000 (Google Scholar, 2024)

Author: Ben Shneiderman, University of Maryland (inventor of the *treemap*, 1991)

Seven Data Types

Type	Example
1-D (linear)	Time series, ranked lists
2-D (planar)	Geographic maps, scatter
3-D (volumetric)	MRI scans, CT volumes
Temporal	Timelines, Gantt charts
Multi-dimensional	Census microdata (10+ vars)
Tree	File system, org chart
Network	Social graph, dependency DAG

The Visual Information Seeking Mantra

“Overview first,
zoom and filter

Seven Tasks (the same for every type)

Shneiderman's Treemap and Dynamic Queries

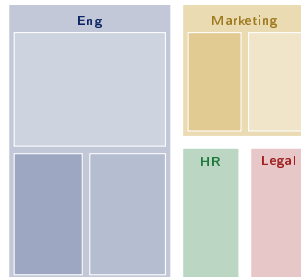
Treemap: Visualising Hierarchies as Area

Ben Shneiderman invented the **treemap** in 1991 (prior to the 1996 paper) to visualise a file system:

- Each file/directory is a rectangle
- **Area** $\propto$ file size
- **Colour** $\propto$ a second variable (modification date, file type)
- Hierarchy encoded by nesting (parent rectangle contains child rectangles)
- Enables inspection of **100,000+ nodes** at once — impossible with a tree diagram

Big Data applications: disk usage analysis (WinDirStat, ncd), stock market heatmaps (Bloomberg, Finviz), NYT budget treemaps, Tableau built-in treemap chart type.

Treemap: disk usage by department



area = storage used; colour = growth rate

Dynamic Queries

Shneiderman proposed **dynamic queries**: sliders and buttons that update the visualisation in

Impact Today — Tufte (1983) & Shneiderman (1996)

How These Works Shape Modern Tools

Concept	Modern instantiation
Data-ink ratio	Tableau's "Remove formatting" one-click; Grafana's minimal theme
Chartjunk removal	D3's default: no axes, no gridlines (add only what you need)
Small multiples	Tableau's "Pages" shelf; Facet wrap in ggplot2; facet_grid in R
Sparklines	Grafana stat panel; Bloomberg terminal; GitHub contribution heatmap

Stanford CS246 / CMU 15-721 Perspective

- Stanford CS246 assigns Tufte as required reading for the **results communication** module — every data mining result is only as good as its visualisation
- CMU 15-721 uses Shneiderman's taxonomy to frame the **query interface design** for OLAP dashboards: a time-series drill-down is a 1-D → temporal transition in Shneiderman's taxonomy
- MIT 6.830 notes that Tufte's data-ink principle directly motivates **columnar storage**: retrieve only the columns ("data ink") needed; ignore the rest ("chartjunk")

Case Study — New York Times Graphics Desk (2010–2022)

The NYT Graphics Desk: Data Journalism at Scale

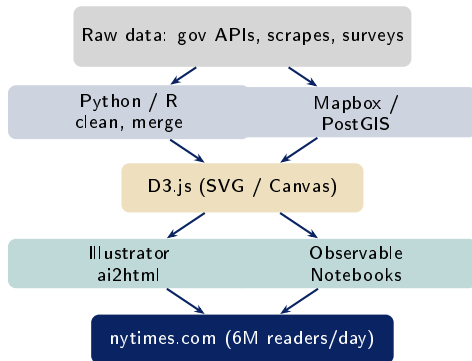
Who They Are and What They Build

The New York Times Graphics Desk is a team of ≈ 50 journalists, designers, and engineers who produce **interactive data visualisations** for nytimes.com — reaching **6 million readers per day**.

Notable members:

- **Mike Bostock** (2010–2015): invented **D3.js** while at NYT; used it to build the 2012 election maps
- **Amanda Cox**: Upshot editor; “You Draw It” participatory journalism pieces
- **Hannah Beachler, Josh Katz**: COVID-19 tracker team (2020–2022)

Their stack: **D3.js** for custom interactive graphics, **R + ggplot2** for exploratory analysis, **Python + pandas** for data processing, **Mapbox** for geographic layers, **ai2html** (Adobe Illustrator → HTML) for static.



Case: NYT COVID-19 Tracker (2020–2022)

The Design Challenge

From March 2020 the NYT Graphics Desk built and maintained the most-read COVID-19 data tracker in the world:

- **Data sources:** Johns Hopkins CSSE, CDC, 50 state health departments, county-level reporting — all with inconsistent schemas, missing values, and retroactive corrections
- **Daily pipeline:** Python scrapers → pandas cleaning → GitHub CSV repository → D3 charts auto-updated every hour
- **Charts produced:** 7-day rolling average line charts (Tufte small multiples for all 50 states), choropleth maps (county-level case rates), vaccination progress bar charts
- **Reach:** tracker page loaded **800 million times** in 2020; cited in CDC, WHO, and UK SAGE policy briefs

Tufte + Shneiderman Principles Applied

- **Data-ink ratio:** no gridlines on case charts; only a light horizontal reference at the prior peak
- **Small multiples:** 50-state chart grid lets readers compare all states simultaneously without animation
- **Overview first:** national total at the top; state breakdown below; county detail on click
- **Zoom and filter:** state selector dropdown (Shneiderman dynamic query); date range slider
- **Details-on-demand:** hover tooltip shows exact date, count, and 7-day change
- **7-day rolling average:** reduces day-of-week reporting artefacts — a data quality decision encoded in the visual design

Key Lessons from the NYT Case

- 1 **Data quality is a visualisation problem:** the NYT's biggest engineering effort was *cleaning* data, not drawing charts; retroactive state corrections forced D3 charts to support negative bars (days where cases were revised down)
- 2 **Accessibility is non-negotiable at 6M readers/day:** every chart had a data table alternative for screen readers; colour scales were tested for colour-blindness (deuteranopia, protanopia)
- 3 **Small multiples beat animation:** the 50-state grid (Tufte small multiples) was more actionable than an animated map of spreading cases
- 4 **Rolling averages are editorial decisions:** choosing 7-day vs. 14-day average changes *what story the chart tells*; these are journalistic choices with ethical weight
- 5 **GitHub as data infrastructure:** publishing the raw CSV on GitHub enabled 1,000+ derivative dashboards worldwide within days

Live Exercise

Live Exercise — D3.js Bar Chart with Tufte Styling

Task (25 minutes, pairs)

Build a minimal D3.js bar chart that follows

Tufte's data-ink principles:

- 1 No gridlines (or only a single light reference line)
- 2 y-axis starts at 0
- 3 No fill gradient — use a flat colour
- 4 Tooltip on hover (Shneiderman: details-on-demand)
- 5 Add a **sparkline** for each bar showing the 7-day trend

Dataset: weekly Kafka message counts for 5 topics (generate 7×5 random values).

Extension: add a dropdown filter to select one topic and zoom into its 7-day line chart.

HTML / D3 Skeleton

```
<!DOCTYPE html>
<meta charset="utf-8">
<style>
  .bar { fill: steelblue; }
  .bar:hover { fill: #b5651d; }
  .axis path, .axis line {
    stroke: #ccc; stroke-width: 0.5px; }
  .tooltip {
    position: absolute; background: #fff;
    border: 1px solid #ccc; padding: 6px;
    font-size: 11px; pointer-events: none; }
</style>
<body>
<script src=
  "https://d3js.org/d3.v7.min.js"></script>
<script>
const topics = ["orders","clicks",
  "errors","logins","search"];
const data = topics.map(t => ({
  topic: t,
  count: Math.floor(Math.random()*50000)
    + 10000
})));
```

Seminal Works

- Tufte, E. R. (1983; 2nd ed. 2001). *The Visual Display of Quantitative Information*. Graphics Press.
- Shneiderman, B. (1996). The eyes have it: A task by data type taxonomy for information visualizations. *Proc. IEEE Visual Languages*, 336–343.

Textbooks [SA], [TE], [LRU]

- **[SA]** Acharya, S., & Chellappan, S. (2015). *Big Data and Analytics* (Ch. 10 — Data Visualisation). Wiley.
- **[TE]** Erl, T., Khattak, W., & Buhler, P. (2016). *Big Data Fundamentals* (Ch. 11 — Visual Analytics). Prentice Hall.
- **[LRU]** Leskovec, J., Rajaraman, A., & Ullman, J. D. (2020). *Mining of Massive Datasets*, 3rd ed. (Ch. 1 — Data and visualisation). Cambridge UP.

Case Study Sources

- Bestack, M., Oriyotale, V., & Heer, J. (2011). D3: Data-driven documents. *IEEE Trans.*

Key Takeaways

- Tufte's **data-ink ratio** is a falsifiable metric: given two charts showing the same data, the one with the higher data-ink ratio communicates more efficiently. Apply it to every dashboard you build.
- Shneiderman's **mantra** (overview → zoom/filter → details) is a universal interaction design pattern — Tableau, Kibana, and D3 implement it differently but all adhere to it
- The NYT COVID tracker proved that **data quality is the invisible half of data visualisation**: the hardest engineering work was schema normalisation across 50 state health departments, not SVG rendering
- **D3.js** (created at NYT) is the direct application

Quote of the Week

“Above all else, show the data.”

— Edward Tufte (1983)

*“Overview first,
zoom and filter,
then details-on-demand.”*

— Ben Shneiderman (1996)

Part 1 Preview — Week 10

Social Media, Time Series & Health Data Analytics